

UNIVERSITY OF THE ARMED FORCES - ESPE

Department of Exact Sciences

Object - Oriented Programming

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WS23 - AnimalFarmPrototype

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Workshop: WS23 – Animal Farm Prototype
Topic: Inheritance and Object-Oriented Programming

WS23 – ANIMAL FARM PROTOTYPE

1. Introduction

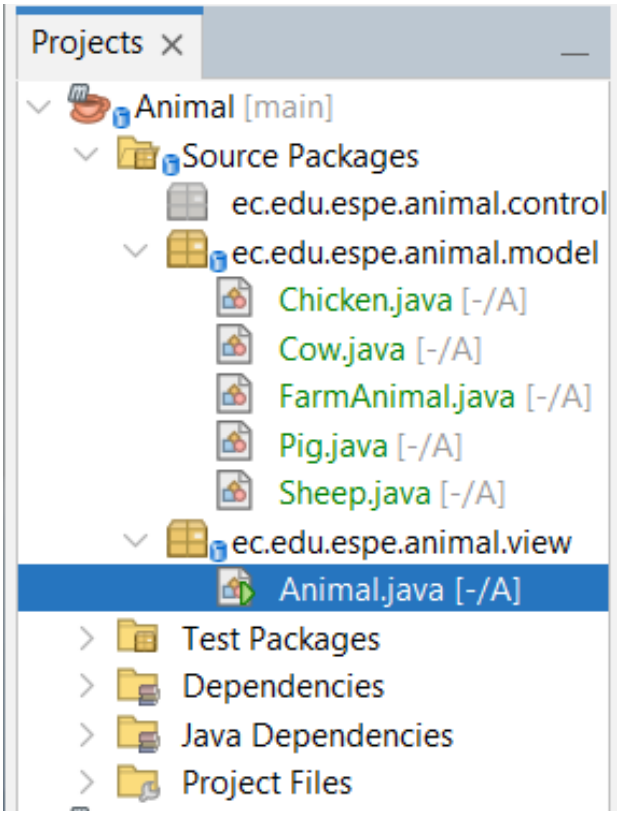
This workshop consists of developing an Animal Farm application using Java, Python, and TypeScript. The program applies object-oriented programming concepts such as inheritance, encapsulation, constructors, and methods.

2. Java Implementation

The first version of the Animal Farm Prototype was developed in Java using Apache NetBeans.

The project includes the `FarmAnimal` parent class and four child classes: `Chicken`, `Cow`, `Pig`, and `Sheep`.

2.1. Project Structure



2.2. Program Execution

The Java application uses an interactive menu to create animals and enter their attributes.

```
========== FARM ANIMAL MENU ==========
1. Create Chicken
2. Create Cow
3. Create Pig
4. Create Sheep
5. Exit
Choose an option: 1
----- ENTER CHICKEN DATA -----
Enter id: 01
Enter bread: Leghorn
Enter born date:
Day: 15
Month: 3
Year: 2026
Is molting? (true/false): false
Number of eggs this month: 25
Is meat? (true/false): false
Chicken -> Chicken{FarmAnimal{id=1, bread=Leghorn, bornOnDate=2026}, isMolting=false, numberOffLatIsMonth=25, isMeat=false}
```

```
========== FARM ANIMAL MENU ==========
1. Create Chicken
2. Create Cow
3. Create Pig
4. Create Sheep
5. Exit
Choose an option: 2
----- ENTER COW DATA -----
Enter id: 02
Enter bread: Holstein
Enter born date:
Day: 10
Month: 6
Year: 2026
Is producing milk? (true/false): true
Liters of milk per week: 150
Cow -> Cow{FarmAnimal{id=2, bread=Holstein, bornOnDate=2026}, isProducingMilk=true, litersOfMilkPerWeek=150.0}
Milking the cow
```

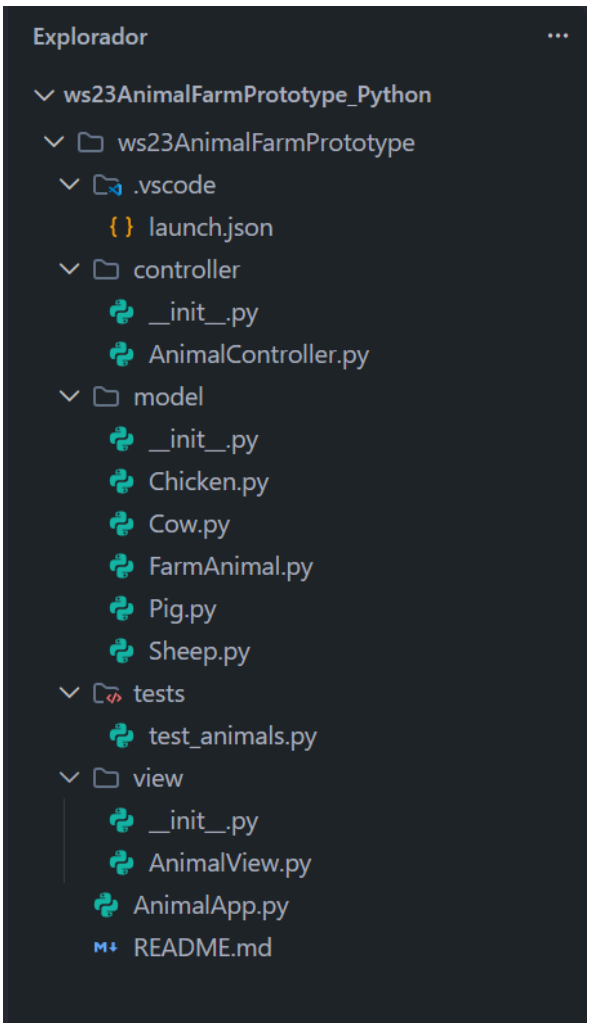
```
========== FARM ANIMAL MENU ==========
1. Create Chicken
2. Create Cow
3. Create Pig
4. Create Sheep
5. Exit
Choose an option: 3
----- ENTER PIG DATA -----
Enter id: 03
Enter bread: Merino
Enter born date:
Day: 20
Month: 4
Year: 2026
Pig -> Pig{FarmAnimal{id=3, bread=Merino, bornOnDate=2026}}
```

```
========== FARM ANIMAL MENU ==========
1. Create Chicken
2. Create Cow
3. Create Pig
4. Create Sheep
5. Exit
Choose an option: 4
----- ENTER SHEEP DATA -----
Enter id: 04
Enter bread: Yorkshire
Enter born date:
Day: 12
Month: 8
Year: 2026
Enter last sheering date:
Day: 5
Month: 6
Year: 2026
Sheep -> Sheep{FarmAnimal{id=4, bread=Yorkshire, bornOnDate=2026}, lastSheering=2026}
```

3. Python Implementation

The second version of the Animal Farm Prototype was developed in Python, maintaining the same object-oriented structure and functionality.

3.1. Project Structure



3.2. Program Execution

The Python application allows users to select an animal, enter its attributes, and display the information.

```
> py AnimalApp.py
========== FARM ANIMAL MENU ==========
1. Create Chicken
2. Create Cow
3. Create Pig
4. Create Sheep
5. Exit
Choose an option: 1
----- ENTER CHICKEN DATA -----
Enter id: 01
Enter bread:
            Leghorn
Enter born date:
Day: 15
Month: 3
Year: 2026
Is molting? (true/false): false
Number of eggs this month: 25
Is meat? (true/false): false
Chicken -> Chicken{FarmAnimal{id=1, bread=Leghorn, bornOnDate=2026}, isMolting=false, numberOffLatIsMonth=25, isMeat=false}
```

```
Choose an option: 2
----- ENTER COW DATA -----
Enter id: 02
Enter bread: Holstein
Enter born date:
Day: 10
Month: 6
Year: 2026
Is producing milk? (true/false): true
Liters of milk per week: 150
Cow -> Cow{FarmAnimal{id=2, bread=Holstein, bornOnDate=2026}, isProducingMilk=true, litersOfMilkPerWeek=150.0}
Milking the cow
```

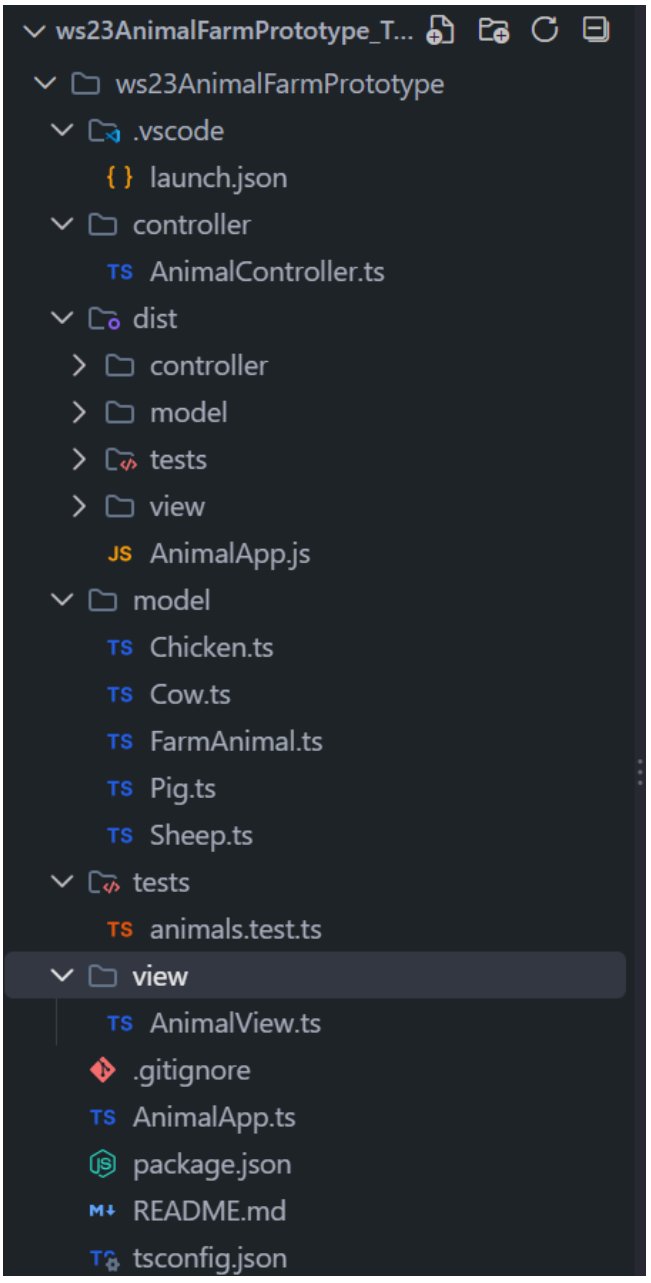
```
Choose an option: 3
----- ENTER PIG DATA -----
Enter id: 03
Enter bread: Merino
Enter born date:
Day: 20
Month: 4
Year: 2026
Pig -> Pig{FarmAnimal{id=3, bread=Merino, bornOnDate=2026}}
```

```
Choose an option: 4
----- ENTER SHEEP DATA -----
Enter id: 04
Enter bread: Merino
Enter born date:
Day: 5
Month: 09
Year: 2026
Enter last sheering date:
Day: 5
Month: 6
Year: 2022
Sheep -> Sheep{FarmAnimal{id=4, bread=Merino, bornOnDate=2026}, lastSheering=2022}
```

4. TypeScript Implementation

The third version of the Animal Farm Prototype was developed in TypeScript, applying inheritance, encapsulation, and modular programming.

4.1. Project Structure



4.2. Program Execution

The TypeScript application provides an interactive menu to register and display farm animals.

```
❯ npm start

> ws23-animal-farm-prototype@1.0.0 start
> node dist/AnimalApp.js

========== FARM ANIMAL MENU ==========
1. Create Chicken
2. Create Cow
3. Create Pig
4. Create Sheep
5. Exit
Choose an option: 1
----- ENTER CHICKEN DATA -----
Enter id: 01
Enter bread: Leghorn
Enter born date:
Day: 15
Month: 3
Year: 2026
Is molting? (true/false): false
Number of eggs this month: 25
Is meat? (true/false): false
Chicken -> Chicken{FarmAnimal{id=1, bread=Leghorn, bornOnDate=2026}, isMolting=false, numberOffLatIsMonth=25, isMeat=false}
```

```
Choose an option: 2
----- ENTER COW DATA -----
Enter id: 02
Enter bread: Holstein
Enter born date:
Day: 10
Month: 6
Year: 2026
Is producing milk? (true/false): true
Liters of milk per week: 150
Cow -> Cow{FarmAnimal{id=2, bread=Holstein, bornOnDate=2026}, isProducingMilk=true, litersOfMilkPerWeek=150}
Milking the cow
```

```
Choose an option: 3
----- ENTER PIG DATA -----
Enter id: 03
Enter bread: Yorkshire
Enter born date:
Day: 5
Month: 6
Year: 2022
Pig -> Pig{FarmAnimal{id=3, bread=Yorkshire, bornOnDate=2022}}
```